

1 Find the linearly independent series solutions $y_1(x)$ and $y_2(x)$ of the differential equation $\frac{d^2 y}{dx^2} + \frac{2}{3x} \frac{dy}{dx} + \frac{x}{3} y = 0$ near $x = 0$ and form the general solution.

2 Given that $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$, solve the following partial differential Equation

$$D^2 (D - 2D') Z = 6x^2 y + \sin(x + 2y).$$

3 Find the linearly independent series solutions $y_1(x)$ and $y_2(x)$ of the differential equation $2 \frac{d^2 y}{dx^2} + \left(\frac{1+x}{x} \right) \frac{dy}{dx} + 3y = 0$ near $x = 0$ and form the general solution.

4. Given that $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$, solve the following partial differential equation

$$(D^2 - 9D'^2) Z = 12x^2 + \sin(3x + 5y).$$

5 Find the linearly independent series solutions $y_1(x)$ and $y_2(x)$ of the differential equation $\frac{d^2 y}{dx^2} + \frac{1}{2x} \frac{dy}{dx} + \frac{1}{4x} y = 0$ near $x = 0$ and form the general solution.

6 .Given that $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$, solve the following partial differential Equation.

$$(D^4 - D'^4) Z = e^{x+y} + x^4 y^4.$$